ENGINEERING CASE STUDY

Stopping Silent PostgreSQL Schema Drift

How I built an automated schema diffing and static linter engine in pure Python.

- ⚡ Zero-connection CI snapshot diffing
- ⚡ Topological DAG constraint ordering
- ⚡ 21 static database health rules

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THE PROBLEM

Schema Drift Is Silent Chaos

Staging matches production until 2 AM on deployment day. One manual **ALTER TABLE** last month silently breaks your ORM migration tonight.

- ✕ Unapplied DDL changes slip into production
- ✕ ORM migrations lock tables unexpectedly
- ✕ Emergency hotfixes create permanent drift

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ARCHITECTURE: STEP 1

Offline Snapshots, Zero Live DBs

You shouldn't need a live database in a CI pipeline just to verify schema drift. SchemaDrift serializes tables, indexes, and foreign keys into lightweight JSON snapshots.

- ✓ Check snapshot JSON directly into Git
- ✓ Run diffs in GitHub Actions with zero secrets
- ✓ Audit changes before opening a pull request

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ARCHITECTURE: STEP 2

Topological DAG Resolution

Naively dropping or adding foreign keys causes immediate deadlocks and constraint violations if dependencies execute out of order.

- ✓ Models table relationships as a directed graph
- ✓ Topological sorting guarantees valid execution order
- ✓ Drops child FKs before altering parent tables

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Deterministic SQL, Always Transactional

Generated migration SQL runs inside an explicit transaction block. If any assertion or constraint check fails, the transaction rolls back cleanly.

- ✓ Wrapped in explicit **BEGIN … COMMIT**
- ✓ Zero half-applied DDL states on failure
- ✓ Full human-readable diff output via CLI

STATIC ANALYSIS

21 Built-In PostgreSQL Rules

Diffing isn't enough if your schema has hidden time bombs. SchemaDrift lints your schema against real production database anti-patterns.

- Unindexed foreign keys (table-lock risks)
- 32-bit integer PKs (2.14B overflow halts)
- Case-sensitive mixedCase column names
- Inexact floats on monetary columns

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SUMMARY

SchemaDrift at a Glance

A complete database safety layer designed for PostgreSQL pipelines.

- **187 Automated Tests:** 98.1% coverage with Pytest
- **21 Linter Rules:** Catching silent database risks
- **Zero Live DB in CI:** Snapshot diffing in Git PRs
- **Automated PR Review:** Listed on GitBot

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Save / Screenshot 📸

GET STARTED

Open Source on GitHub

Try SchemaDrift on your PostgreSQL databases or wire it into your CI pipelines today.

github.com/Asadshah7950/schemadrift

CLI tool · Python package · GitBot reviewer

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